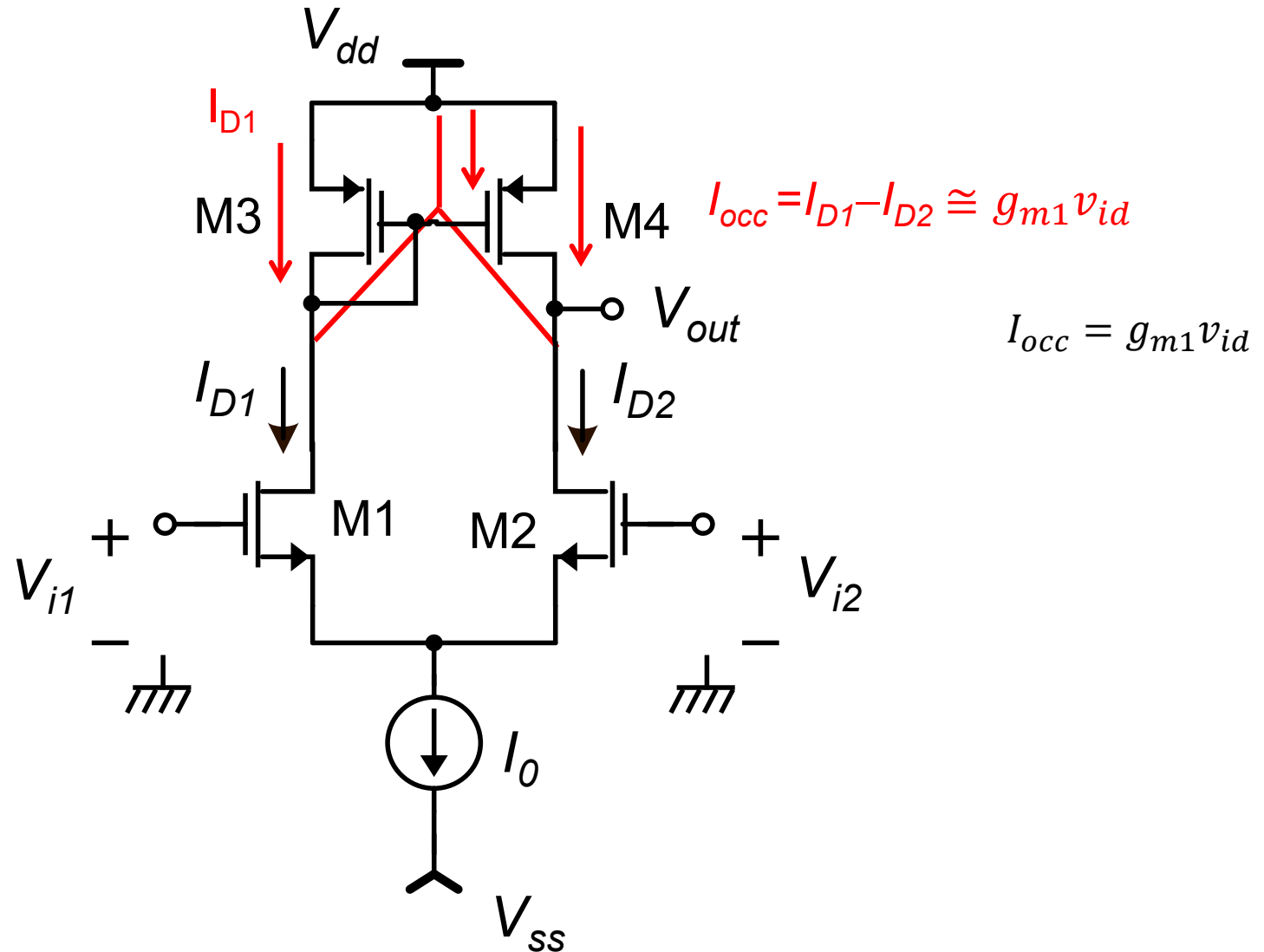


Differential Amplifiers

CMOS differential amplifier with current mirror load

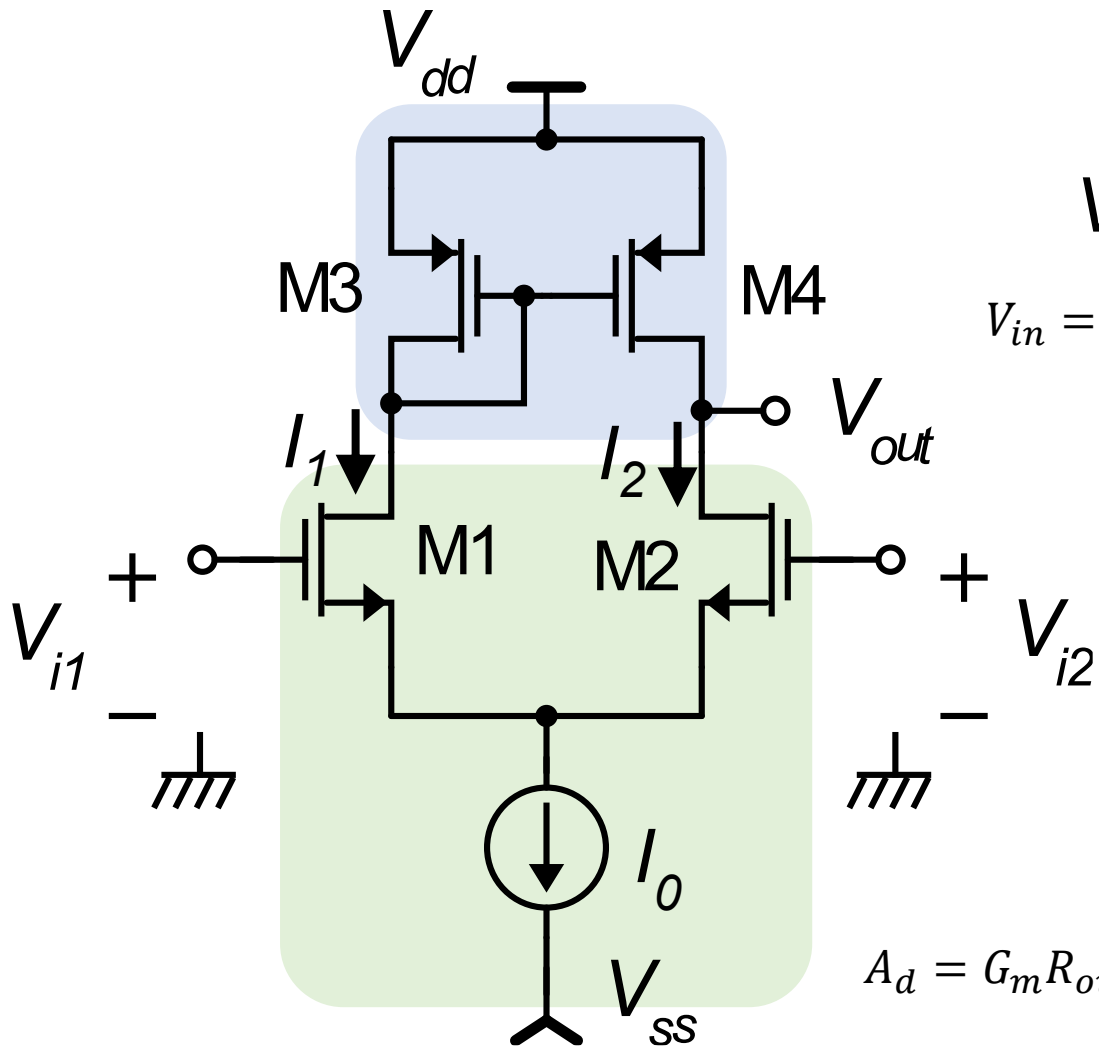
Specifications

- We need a S/E output port
- High CMRR (> 80 dB)
- High gain (~ 40 dB) even at low supply voltages ($V_{dd} - V_{ss}$).

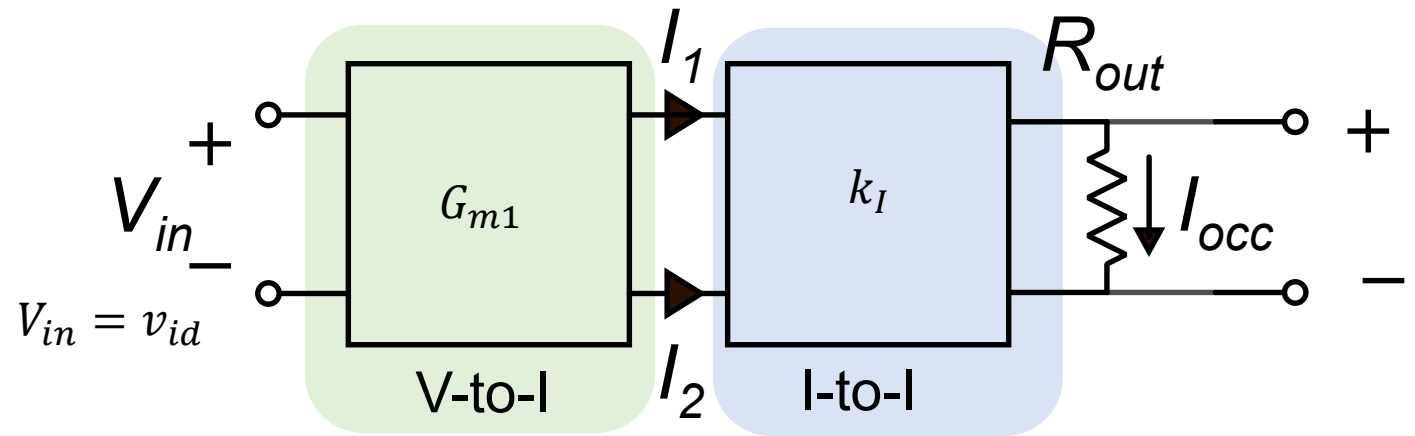


This differential amplifier has single-ended output: V_{out} !

Subunits of the amplifier with mirror load



$$A_d = G_m R_{out} = g_{m1} R_{out}$$



$$(I_1 - I_2) = G_{m1} V_{in}$$

$$I_{occ} = k_I (I_1 - I_2)$$

$$G_{m1} = g_{m1} (= g_{m2})$$

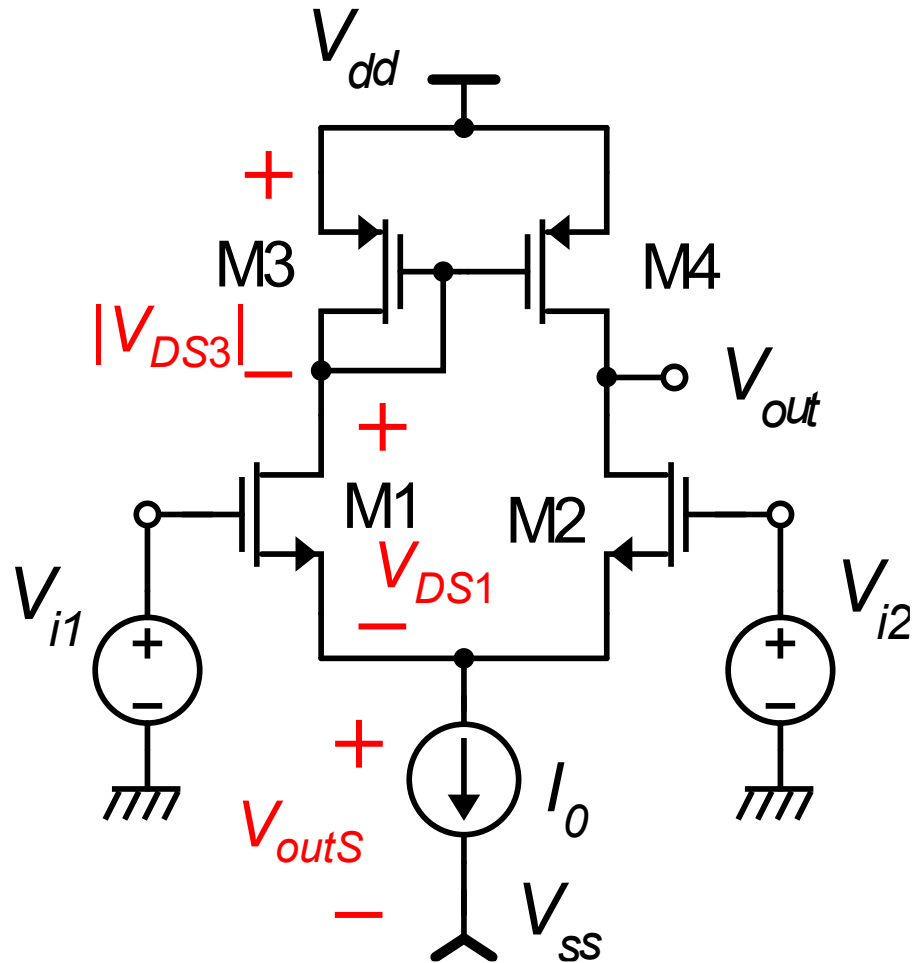
$$k_I = 1$$

$$I_{occ} = G_m V_{in}$$

$$G_m = G_{m1} k_I = g_{m1}$$

This differential amplifier has single-ended output: V_{out} !

Minimum supply voltage $V_{dd} - V_{ss}$



$$V_{dd} - V_{ss} = V_{outS} + V_{DS1} + |V_{DS3}|$$

$$V_{DS3} = V_{GS3}$$

$$\min(V_{outS}) = V_{MIN}$$

$$\min(V_{dd} - V_{ss}) = V_{MIN} + V_{DSAT1} + |V_{GS3}|$$

Example: $V_{MIN} = 100 \text{ mV}$

$V_{DSAT2} = 100 \text{ mV}$

$|V_{GS3}| = 0.5 \text{ V}$

$$\min(V_{dd} - V_{ss}) \cong 0.7 \text{ V}$$

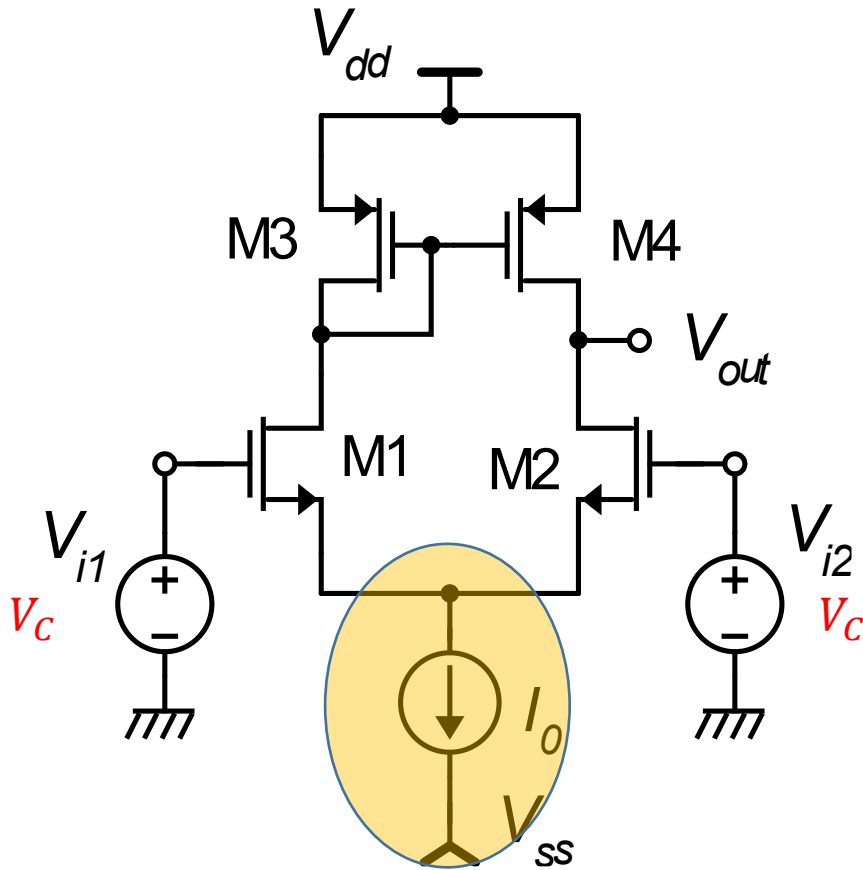
This differential amplifier has single-ended output: V_{out} !

Input common mode range

Lower limit

As V_C is progressively decreased, also V_S decreases at the same pace and eventually the voltage across the current source I_0 will get smaller than the minimum value V_{MIN} . From that point on, I_0 will rapidly decrease, turning off the stage.

$$\min(V_C) = V_{MIN} + V_{GS1}$$



This differential amplifier has single-ended output: V_{out} !

Input common mode range

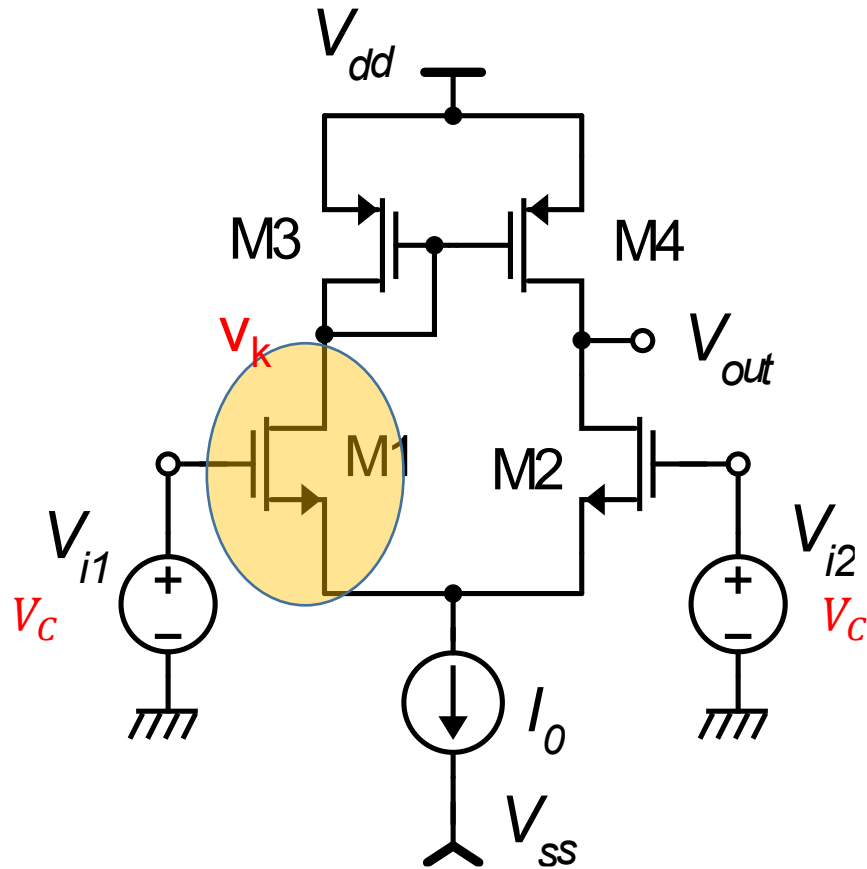
Upper limit

As V_C is progressively increased, also V_S increases at the same pace. Since $V_{out} = V_K = V_{D2} = V_{D1}$ is fixed, eventually V_{DS1} and V_{DS2} will drop below the saturation voltage.

$$V_{DS1} = V_K - V_{S1} \geq V_{DSAT1}$$

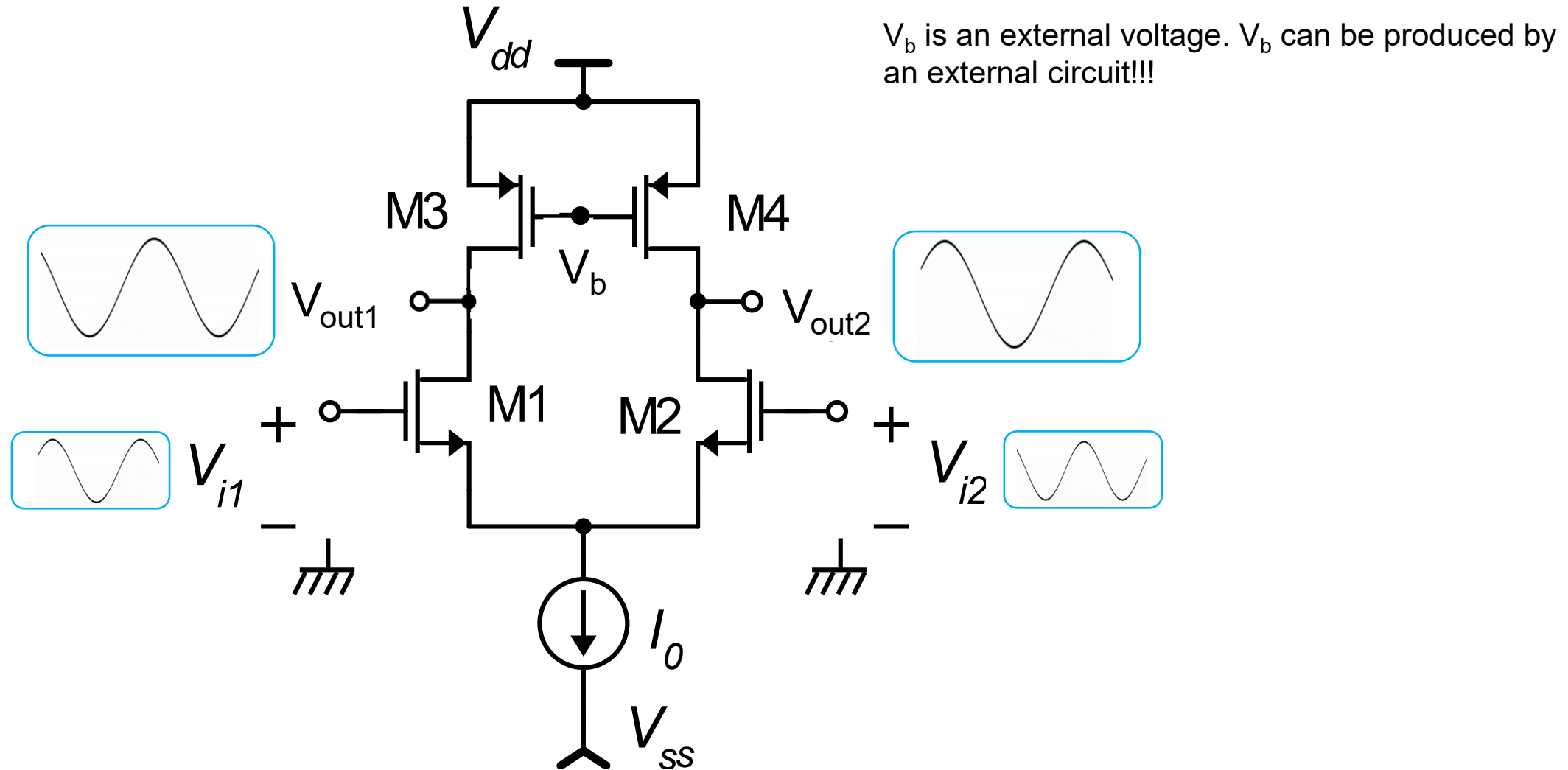
$$\overbrace{V_{dd} - |V_{GS3}|} - \overbrace{(V_C - V_{GS1})} \geq V_{DSAT1}$$

$$V_{dd} - |V_{GS3}| + V_{GS1} - V_{DSAT1} \geq V_C$$



This differential amplifier has single-ended output: V_{out} !

Fully Differential Amplifier



This differential amplifier has **differential outputs**: V_{out1} and V_{out2} !

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.